RAJU MANDAL

Java Full Stack Developer | Java 17 | Spring Boot | Microservices | Angular | AWS | GCP

Faridabad, Haryana, India | Phone: +91-8851210791 | Email: rajukumarmandal197@gmail.com

LinkedIn: linkedin.com/in/raju-mandal

PROFESSIONAL SUMMARY

Java Full Stack Developer with **6+ years of experience** building secure, scalable, cloud-native enterprise applications using **Java 8/17, Spring Boot, Spring Cloud and Angular**. Strong in **Microservices architecture, event-driven systems (GCP Pub/Sub, Apache Kafka, RabbitMQ), RESTful API design, Spring Security with JWT/OAuth 2.0, multi-tenant RBAC, Spring Data JPA/Hibernate, PostgreSQL and MongoDB**. Built a low-code workflow and document-automation platform (orchestration, Gemini/LLM integration, rule engine, API gateway, IAM) and healthcare and background-verification systems. Hands-on with **AWS, GCP, Docker, Kubernetes, CI/CD** and Agile/Scrum across the full SDLC.

CORE COMPETENCIES

Java Full Stack Development | Microservices Architecture | Event-Driven Architecture | RESTful API Design | Spring Boot & Spring Cloud | API Gateway | Multi-Tenant SaaS | RBAC | Workflow Orchestration | LLM / Gemini API Integration | Multithreading & Concurrency | System Design | Cloud-Native Deployment (AWS & GCP) | Agile/Scrum

TECHNICAL SKILLS

Frontend: Angular, TypeScript, JavaScript, RxJS, HTML5, CSS3, Bootstrap, Tailwind CSS, Kendo UI

Backend: Java 8/17, Spring Boot, Spring MVC, Spring Cloud Gateway, Spring Cloud GCP, Spring Data JPA, Hibernate, Spring WebSocket (STOMP), REST APIs, Microservices, SOAP, Resilience4j, Maven, Gradle

Security: Spring Security, JWT (Access & Refresh Tokens), OAuth 2.0, RBAC, BCrypt, XSS & SQL Injection Protection

Databases: PostgreSQL, MongoDB (Change Streams), MySQL, MS SQL Server, SQL, Redis

Messaging: GCP Pub/Sub, Apache Kafka, RabbitMQ, WebSockets, Transactional Outbox Pattern

Java Core & Concurrency: Collections, Streams API, Lambdas, ExecutorService, CompletableFuture, Multithreading

Cloud: AWS (EC2, S3, Lambda, RDS, SQS, SNS, EKS, CloudWatch, IAM), GCP (Cloud Run, Pub/Sub, Cloud Storage, Cloud Logging, Cloud Monitoring)

Testing & Tools: JUnit 5, Mockito, Spring Boot Test, Jasmine, Karma, Postman, Swagger/OpenAPI, IntelliJ IDEA, Jira

DevOps: Docker, Kubernetes, Git, CI/CD Pipelines

Concepts: DSA, OOP, SOLID, Design Patterns (Strategy, Factory, Builder), Circuit Breaker, Clean Code, SDLC

PROFESSIONAL EXPERIENCE

Technical Consultant *Mar 2024 - Sep 2026*

Digivate Labs Pvt Ltd, Gurgaon

Anota - Low-Code Workflow & Document Automation Platform

Use-case-agnostic, multi-tenant low-code platform where each client configures its own workflow steps (e.g. upload -> classification -> extraction -> rule validation -> review) and pushes cases through them; built on Java/Spring Boot microservices with an Angular builder UI.

- Built **Spring Boot microservices** for a **configurable workflow engine** and delivered the end-to-end **health-insurance claim flow**: multi-mode upload APIs, document classification, OCR-backed extraction, human correction, multi-level review, third-party checks (red flag, hospital-bill validation, claim-amount calculation) and PDF settlement-letter generation.
- Re-architected the event pipeline from a **MongoDB Change Streams** watcher to **GCP Pub/Sub** using **Spring Cloud GCP**, with per-step success/failure topics; built the **Relay Service** that auto-provisions topics and subscriptions for each configured step and the **Mapper Service** that builds every worker's request from case data plus configuration.
- Built Java **workers** for classification (sync, async and batch), extraction, splitter, page numbering, rule, notification and third-party integrations, using **ExecutorService and CompletableFuture** for concurrent message processing.
- Developed the **API-Facade AI orchestration service** called by all workers to reach **Google Gemini** and in-house ML/LLM APIs; added **Resilience4j retry, circuit breaker and timeouts** for unavailable or resource-exhausted model endpoints, and used the **Strategy pattern** behind Java interfaces to switch between sync and async execution.
- Built a configurable **Rule Engine** that evaluates client-defined conditions and returns a boolean plus a business-readable message (e.g. policy-maturity checks) to drive workflow branching.

- Solved the "DB updated but publish failed" consistency gap after the Pub/Sub move with a **Transactional Outbox** approach and idempotent consumers, so no workflow step was lost or processed twice.
- Migrated transactional data from **MongoDB to PostgreSQL** using **Spring Data JPA/Hibernate** to reduce cost, enforce schema and cut the connection load caused by change streams across all workflows.
- Implemented **IAM with Spring Security**: JWT **access and refresh tokens** carrying tenant ID, use case and role claims, enabling **multi-tenant isolation and RBAC** in every downstream service; passwords stored with BCrypt hashing.
- Built the **API Gateway on Spring Cloud Gateway** through which all microservice APIs are routed, with global filters for authentication and **XSS/script and SQL-injection protection**.
- Developed the **"Liquid" dynamic CRUD service** (4 generic REST APIs for any collection), cloud-agnostic upload storage (**AWS S3, GCP Cloud Storage**), an **Email/SMS Notification Service**, and live status updates over **WebSockets (STOMP)**.
- Developed the Angular **UI Builder / Form Builder** (drag-and-drop pages such as document upload and case-status grids) and the **JSON Mapper** library, with Spring Boot APIs persisting form and mapping configurations.
- Migrated services from GCP VMs to **Cloud Run** with **Docker** and **CI/CD pipelines**; added **Redis caching** and **Cloud Logging/Monitoring**.
- Wrote unit tests with **JUnit 5 and Mockito** (interface-based mocks) and **Jasmine/Karma**; as a senior developer broke features into Jira tickets, assigned work and reviewed pull requests for new joiners.

Tech Stack: Java 17, Spring Boot, Spring Cloud Gateway, Spring Security, Spring Data JPA, Hibernate, GCP Pub/Sub, Resilience4j, PostgreSQL, MongoDB, Redis, Gemini API, Angular, Docker, Cloud Run, JUnit 5, Mockito

Health Automation, Extraction & Validation (Healthcare Domain)

Healthcare document extraction and validation platform using LLM-based extraction on unstructured clinical documents.

- Designed and implemented **Spring Boot REST APIs** for LLM-based data extraction, validation and persistence, with request validation and global exception handling (**@ControllerAdvice**).
- Implemented **JWT-based authentication and authorization with Spring Security** to secure application endpoints.
- Designed a **multi-level role hierarchy and approval workflow**, ensuring data is written to the database only after approval.
- Built **Angular UI** for viewing, editing, and validating extracted data with role-based access control.
- Deployed applications using **CI/CD pipelines** for consistent, reliable releases.

Tech Stack: Java, Spring Boot, Spring Security, MongoDB, JWT, Angular, TypeScript, CI/CD, Git

Associate Technical Consultant *Feb 2019 - Mar 2024*

Grid Infocom Pvt Ltd, Gurgaon

Peregrine Application (Client: First Advantage, USA)

- Developed an enterprise **Angular application** for background-verification services.
- Integrated **REST and SOAP APIs** and improved mobile responsiveness across the platform.
- Supported production issues and enhancements, ensuring high system reliability.

Peregrine Microservices

- Developed **Spring Boot microservices** with **Spring Data JPA and PostgreSQL** for high-concurrency background-verification workflows.
- Implemented **OAuth 2.0 and JWT** for secure inter-service communication.
- Integrated **Apache Kafka and RabbitMQ** for asynchronous message processing between services.

CERTIFICATIONS

- Java - Aricent & NASSCOM
- Angular, JavaScript, CSS - Udemy
- MongoDB M001 - MongoDB University

EDUCATION

B.Tech in Computer Science, IPU Delhi | 2014 - 2018 | 74%